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1-Phase Half Wave Phase Controlled Rectifier

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Comprehensive Class Notes with Complete Derivations

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1 Module Overview & Lecture Structure

The main aim of the present note is to derive, step by step, all performance expressions of the single-phase half wave phase controlled rectifier. Four load types are considered, say R, RL, RE and RLE. Each load is analysed without and with free wheeling diode (FD). For every case the average output voltage, RMS output voltage, average and RMS values of input and output current, PIV, form factor and ripple factor are obtained from first principles.

Definition 1.1 (Module Philosophy): *A phase controlled rectifier converts fixed AC voltage into variable DC voltage by delaying the turn-on instant of the thyristor by the firing angle α . Every result in this note follows from two operations only: (i) writing KVL for the conducting interval and (ii) integrating over one full period 2π .*

1.1 Lecture Learning Structure

Table 1: 1-Phase Half Wave Phase Controlled Rectifier - Lecture Structure

Lecture	Topic	Duration	Learning Activities
1	Circuit, angles and performance definitions	50 min	Waveform sketching
2	R load without and with FD	50 min	Derivation on board
3	RL load without FD	50 min	Solving first order ODE
4	RL load with FD	50 min	Mode wise analysis
5	RE load without and with FD	50 min	Battery charger problem
6	RLE load without and with FD	50 min	Mode wise analysis
7	Source inductance and overlap angle	50 min	Numerical workshop
8	Numerical determination of β by NR method	50 min	Iteration by calculator

2 Circuit Description and Basic Definitions

2.1 Circuit and Assumptions

A single thyristor T is connected in series with the load across a single-phase AC source. For the FD cases, a diode FD is connected across the load with its cathode at the positive load terminal.

Source voltage:

$$v_s = V_m \sin \omega t, \quad V_m = \sqrt{2} V_s \quad (1)$$

Assumptions:

- Thyristor and diode are ideal (zero drop in ON state, zero leakage in OFF state)
- Source is ideal, except in Section 8 where source inductance L_s is included
- Steady state operation with period 2π in ωt
- One gate pulse is applied at $\omega t = \alpha$ in every cycle

2.2 Angle Definitions

Definition 2.1 (Angles Used in This Note):

- α = firing (delay) angle, measured from the positive going zero crossing of v_s
- β = extinction angle, the value of ωt at which load current falls to zero
- $\gamma = \beta - \alpha$ = conduction angle
- $\theta = \tan^{-1} \left(\frac{\omega L}{R} \right)$ = load impedance angle
- μ = overlap (commutation) angle due to source inductance L_s
- $\delta = \sin^{-1} \left(\frac{E}{V_m} \right)$ = angle at which v_s equals the load EMF E

Load impedance relations:

$$Z = \sqrt{R^2 + (\omega L)^2} \quad (2)$$

$$R = Z \cos \theta, \quad \omega L = Z \sin \theta, \quad \frac{R}{\omega L} = \frac{1}{\tan \theta} \quad (3)$$

2.3 Performance Parameter Definitions

For any periodic output voltage $v_o(\omega t)$ and output current $i_o(\omega t)$ of period 2π :

Average (DC) values:

$$V_o = \frac{1}{2\pi} \int_0^{2\pi} v_o d(\omega t), \quad I_o = \frac{1}{2\pi} \int_0^{2\pi} i_o d(\omega t) \quad (4)$$

RMS values:

$$V_{or} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} v_o^2 d(\omega t)}, \quad I_{or} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} i_o^2 d(\omega t)} \quad (5)$$

Form factor:

$$\text{FF} = \frac{V_{or}}{V_o} \quad (6)$$

Ripple factor:

The output voltage contains a DC part V_o and an AC (ripple) part of RMS value V_{ac} . These two parts are orthogonal over one period. Therefore:

$$V_{or}^2 = V_o^2 + V_{ac}^2 \quad (7)$$

$$V_{ac} = \sqrt{V_{or}^2 - V_o^2} \quad (8)$$

$$\text{RF} = \frac{V_{ac}}{V_o} = \sqrt{\left(\frac{V_{or}}{V_o} \right)^2 - 1} = \sqrt{\text{FF}^2 - 1} \quad (9)$$

The same definitions hold for current, with I_{or} and I_o in place of V_{or} and V_o .

Peak inverse voltage (PIV): the maximum reverse voltage that appears across the device in OFF state.

Input (source) current: without FD the thyristor is in series with the load, so $i_s = i_T = i_o$ at every instant. With FD, $i_s = i_T$ and $i_o = i_T + i_{FD}$.

Input power factor:

$$\text{pf} = \frac{\text{Power delivered to load}}{V_s I_{s,\text{rms}}} \quad (10)$$

3 R Load

3.1 R Load without FD

3.1.1 Operation

- $0 < \omega t < \alpha$: T is forward biased but not fired. $v_o = 0$, $i_o = 0$, $v_T = v_s$
- $\alpha \leq \omega t \leq \pi$: T conducts. $v_o = v_s$, $i_o = v_s/R$, $v_T = 0$
- $\pi < \omega t < 2\pi$: current reaches zero at $\omega t = \pi$ and T turns OFF. $v_o = 0$, $v_T = v_s < 0$

Hence $\beta = \pi$ and $\gamma = \pi - \alpha$.

3.1.2 Average Output Voltage

$$V_o = \frac{1}{2\pi} \int_{\alpha}^{\pi} V_m \sin \omega t d(\omega t) \quad (11)$$

$$= \frac{V_m}{2\pi} \left[-\cos \omega t \right]_{\alpha}^{\pi} \quad (12)$$

$$= \frac{V_m}{2\pi} (-\cos \pi + \cos \alpha) \quad (13)$$

$$V_o = \frac{V_m}{2\pi} (1 + \cos \alpha) \quad (14)$$

At $\alpha = 0$, $V_o = V_m/\pi$ (maximum). At $\alpha = \pi$, $V_o = 0$.

3.1.3 RMS Output Voltage

$$V_{or}^2 = \frac{1}{2\pi} \int_{\alpha}^{\pi} V_m^2 \sin^2 \omega t d(\omega t) \quad (15)$$

$$= \frac{V_m^2}{2\pi} \int_{\alpha}^{\pi} \frac{1 - \cos 2\omega t}{2} d(\omega t) \quad (16)$$

$$= \frac{V_m^2}{4\pi} \left[\omega t - \frac{\sin 2\omega t}{2} \right]_{\alpha}^{\pi} \quad (17)$$

$$= \frac{V_m^2}{4\pi} \left[(\pi - \alpha) + \frac{\sin 2\alpha}{2} \right] \quad (18)$$

$$V_{or} = \frac{V_m}{2\sqrt{\pi}} \sqrt{(\pi - \alpha) + \frac{1}{2} \sin 2\alpha} \quad (19)$$

At $\alpha = 0$, $V_{or} = V_m/2$.

3.1.4 Output and Input Currents

Since $i_o = v_o/R$ at every instant:

$$I_o = \frac{V_o}{R} = \frac{V_m}{2\pi R} (1 + \cos \alpha) \quad (20)$$

$$I_{or} = \frac{V_{or}}{R} = \frac{V_m}{2R\sqrt{\pi}} \sqrt{(\pi - \alpha) + \frac{1}{2} \sin 2\alpha} \quad (21)$$

Source, thyristor and load carry the same current:

$$I_{s,avg} = I_{T,avg} = I_o, \quad I_{s,rms} = I_{T,rms} = I_{or} \quad (22)$$

3.1.5 Form Factor and Ripple Factor

$$\text{FF} = \frac{V_{or}}{V_o} = \frac{\sqrt{\pi} \sqrt{(\pi - \alpha) + \frac{1}{2} \sin 2\alpha}}{1 + \cos \alpha} \quad (23)$$

$$\text{RF} = \sqrt{\text{FF}^2 - 1} \quad (24)$$

At $\alpha = 0$: $\text{FF} = \pi/2 = 1.571$ and $\text{RF} = 1.211$. For R load the current FF and RF are equal to the voltage FF and RF.

3.1.6 PIV and Other Results

PIV: for $\pi < \omega t < 2\pi$, $v_T = v_s$. The most negative value occurs at $\omega t = 3\pi/2$.

$$\text{PIV} = V_m \quad (25)$$

Peak forward blocking voltage: $V_m \sin \alpha$ for $\alpha \leq \pi/2$ and V_m for $\alpha > \pi/2$.

Load power and input power factor:

$$P = I_{or}^2 R = \frac{V_{or}^2}{R} \quad (26)$$

$$\text{pf} = \frac{V_{or}^2/R}{V_s \cdot V_{or}/R} = \frac{V_{or}}{V_s} = \frac{1}{\sqrt{2}} \sqrt{\frac{1}{\pi} \left[(\pi - \alpha) + \frac{1}{2} \sin 2\alpha \right]} \quad (27)$$

Rectification efficiency:

$$\eta = \frac{V_o I_o}{V_{or} I_{or}} = \frac{1}{\text{FF}^2} \quad (28)$$

At $\alpha = 0$, $\eta = 4/\pi^2 = 40.53\%$.

Circuit turn-off time: T is reverse biased from $\omega t = \pi$ to 2π , so $t_c = \pi/\omega$.

3.2 R Load with FD

With R load, $v_o = R i_o \geq 0$ at every instant. The FD is therefore never forward biased and never conducts. All expressions of R load without FD remain valid:

$$V_o = \frac{V_m}{2\pi} (1 + \cos \alpha), \quad V_{or} = \frac{V_m}{2\sqrt{\pi}} \sqrt{(\pi - \alpha) + \frac{1}{2} \sin 2\alpha}, \quad \text{PIV} = V_m \quad (29)$$

The FD blocks v_o during $\alpha \leq \omega t \leq \pi$. Its PIV is V_m for $\alpha \leq \pi/2$ and $V_m \sin \alpha$ for $\alpha > \pi/2$.

Example 3.1 (R Load):

A 1-phase half wave controlled rectifier feeds $R = 10 \Omega$ from 230 V, 50 Hz supply at $\alpha = 60^\circ$.

Peak voltage:

$$V_m = \sqrt{2} \times 230 = 325.27 \text{ V} \quad (30)$$

Average output voltage and current:

$$V_o = \frac{325.27}{2\pi} (1 + \cos 60^\circ) = 51.768 \times 1.5 = 77.65 \text{ V} \quad (31)$$

$$I_o = \frac{77.65}{10} = 7.765 \text{ A} \quad (32)$$

RMS output voltage and current:

$$V_{or} = \frac{325.27}{2} \sqrt{\frac{1}{\pi} \left(\pi - \frac{\pi}{3} + \frac{\sin 120^\circ}{2} \right)} = 162.63 \sqrt{\frac{2.0944 + 0.4330}{3.1416}} \quad (33)$$

$$= 162.63 \times 0.8969 = 145.87 \text{ V} \quad (34)$$

$$I_{or} = \frac{145.87}{10} = 14.587 \text{ A} \quad (35)$$

Form factor and ripple factor:

$$\text{FF} = \frac{145.87}{77.65} = 1.879 \quad (36)$$

$$\text{RF} = \sqrt{1.879^2 - 1} = 1.590 \quad (37)$$

Other results:

$$\text{PIV} = 325.27 \text{ V} \quad (38)$$

$$P = \frac{145.87^2}{10} = 2127.8 \text{ W} \quad (39)$$

$$\text{pf} = \frac{145.87}{230} = 0.634 \text{ (lag)} \quad (40)$$

$$I_{s,\text{avg}} = 7.765 \text{ A}, \quad I_{s,\text{rms}} = 14.587 \text{ A} \quad (41)$$

4 RL Load without FD

4.1 Operation

The inductance opposes the change of current. The current rises slowly from zero at $\omega t = \alpha$ and continues to flow beyond $\omega t = \pi$. T remains ON till the current falls to zero at $\omega t = \beta > \pi$. During $\pi < \omega t < \beta$ the output voltage is negative.

- $\alpha \leq \omega t \leq \beta$: T conducts, $v_o = v_s$
- $\beta < \omega t < 2\pi + \alpha$: T is OFF, $v_o = 0$, $i_o = 0$, $v_T = v_s$

4.2 Load Current Expression

Step 1: KVL during conduction

$$L \frac{di_o}{dt} + R i_o = V_m \sin \omega t, \quad \alpha \leq \omega t \leq \beta \quad (42)$$

Step 2: steady state (forced) component

$$i_{ss} = \frac{V_m}{Z} \sin(\omega t - \theta) \quad (43)$$

Step 3: transient (natural) component

$$i_{tr} = A e^{-\frac{R}{L}t} = A e^{-\frac{\omega t}{\tan \theta}} \quad (44)$$

Step 4: complete solution

$$i_o = \frac{V_m}{Z} \sin(\omega t - \theta) + A e^{-\frac{\omega t}{\tan \theta}} \quad (45)$$

Step 5: initial condition $i_o = 0$ at $\omega t = \alpha$

$$0 = \frac{V_m}{Z} \sin(\alpha - \theta) + A e^{-\frac{\alpha}{\tan \theta}} \quad (46)$$

$$A = -\frac{V_m}{Z} \sin(\alpha - \theta) e^{\frac{\alpha}{\tan \theta}} \quad (47)$$

Step 6: final expression

$$i_o(\omega t) = \frac{V_m}{Z} \left[\sin(\omega t - \theta) - \sin(\alpha - \theta) e^{-\frac{\omega t - \alpha}{\tan \theta}} \right], \quad \alpha \leq \omega t \leq \beta \quad (48)$$

4.3 Extinction Angle

Putting $i_o = 0$ at $\omega t = \beta$ in (48):

$$\sin(\beta - \theta) = \sin(\alpha - \theta) e^{-\frac{\beta - \alpha}{\tan \theta}} \quad (49)$$

This equation is transcendental. It is solved for β numerically (say by Newton-Raphson method, Section 12) for known α and θ . Then $\gamma = \beta - \alpha$.

4.4 Average Output Voltage

$$V_o = \frac{1}{2\pi} \int_{\alpha}^{\beta} V_m \sin \omega t d(\omega t) \quad (50)$$

$$= \frac{V_m}{2\pi} \left[-\cos \omega t \right]_{\alpha}^{\beta} \quad (51)$$

$$V_o = \frac{V_m}{2\pi} (\cos \alpha - \cos \beta) \quad (52)$$

Since $\beta > \pi$, $\cos \beta > -1$. Hence V_o is lower than that of R load for the same α .

4.5 RMS Output Voltage

$$V_{or}^2 = \frac{1}{2\pi} \int_{\alpha}^{\beta} V_m^2 \sin^2 \omega t d(\omega t) \quad (53)$$

$$= \frac{V_m^2}{4\pi} \left[\omega t - \frac{\sin 2\omega t}{2} \right]_{\alpha}^{\beta} \quad (54)$$

$$= \frac{V_m^2}{4\pi} \left[(\beta - \alpha) - \frac{1}{2} (\sin 2\beta - \sin 2\alpha) \right] \quad (55)$$

$$V_{or} = \frac{V_m}{2\sqrt{\pi}} \sqrt{(\beta - \alpha) - \frac{1}{2} (\sin 2\beta - \sin 2\alpha)} \quad (56)$$

4.6 Average Output Current

Method 1: from average voltage. Taking the average of $v_o = Ri_o + L di_o/dt$ over one period:

$$V_o = RI_o + \frac{\omega L}{2\pi} \int_{\alpha}^{\beta} \frac{di_o}{d(\omega t)} d(\omega t) = RI_o + \frac{\omega L}{2\pi} [i_o(\beta) - i_o(\alpha)] \quad (57)$$

Since $i_o(\alpha) = i_o(\beta) = 0$, the average inductor voltage is zero. Therefore:

$$I_o = \frac{V_o}{R} = \frac{V_m}{2\pi R} (\cos \alpha - \cos \beta) \quad (58)$$

Method 2: direct integration (verification).

$$I_o = \frac{1}{2\pi} \int_{\alpha}^{\beta} i_o d(\omega t) \quad (59)$$

$$= \frac{V_m}{2\pi Z} \left[\cos(\alpha - \theta) - \cos(\beta - \theta) - \tan \theta \sin(\alpha - \theta) \left(1 - e^{-\frac{\gamma}{\tan \theta}} \right) \right] \quad (60)$$

Using (49), $\sin(\alpha - \theta) e^{-\gamma/\tan \theta} = \sin(\beta - \theta)$. Then:

$$I_o = \frac{V_m}{2\pi Z} [\{\cos(\alpha - \theta) - \tan \theta \sin(\alpha - \theta)\} - \{\cos(\beta - \theta) - \tan \theta \sin(\beta - \theta)\}] \quad (61)$$

$$= \frac{V_m}{2\pi Z} \left[\frac{\cos \alpha}{\cos \theta} - \frac{\cos \beta}{\cos \theta} \right] = \frac{V_m}{2\pi R} (\cos \alpha - \cos \beta) \quad (62)$$

Both methods give the same result.

4.7 RMS Output Current

$$I_{or}^2 = \frac{1}{2\pi} \int_{\alpha}^{\beta} i_o^2 d(\omega t) \quad (63)$$

Let $K = \sin(\alpha - \theta)$. Squaring (48) gives three integrals:

$$J_1 = \int_{\alpha}^{\beta} \sin^2(\omega t - \theta) d(\omega t) = \frac{\gamma}{2} - \frac{1}{4} [\sin 2(\beta - \theta) - \sin 2(\alpha - \theta)] \quad (64)$$

$$J_2 = \int_{\alpha}^{\beta} K^2 e^{-\frac{2(\omega t - \alpha)}{\tan \theta}} d(\omega t) = \frac{K^2 \tan \theta}{2} \left(1 - e^{-\frac{2\gamma}{\tan \theta}} \right) \quad (65)$$

$$J_3 = \int_{\alpha}^{\beta} \sin(\omega t - \theta) e^{-\frac{\omega t - \alpha}{\tan \theta}} d(\omega t) = \sin \theta \left(\sin \alpha - e^{-\frac{\gamma}{\tan \theta}} \sin \beta \right) \quad (66)$$

Therefore:

$$I_{or} = \frac{V_m}{Z} \sqrt{\frac{1}{2\pi} (J_1 + J_2 - 2KJ_3)} \quad (67)$$

Check: for $L = 0$, $\theta = 0$, $\tan \theta = 0$ and $\beta = \pi$. Then $J_2 = J_3 = 0$ and $J_1 = \frac{1}{2}(\pi - \alpha) + \frac{1}{4} \sin 2\alpha$, which is the R load result.

4.8 Input Current, FF, RF and PIV

Input current: T is in series with the load.

$$I_{s,avg} = I_{T,avg} = I_o, \quad I_{s,rms} = I_{T,rms} = I_{or} \quad (68)$$

Form factor and ripple factor of output voltage:

$$FF = \frac{V_{or}}{V_o} = \frac{\sqrt{\pi} \sqrt{(\beta - \alpha) - \frac{1}{2}(\sin 2\beta - \sin 2\alpha)}}{\cos \alpha - \cos \beta} \quad (69)$$

$$RF = \sqrt{FF^2 - 1} \quad (70)$$

Current ripple factor:

$$RF_i = \sqrt{\left(\frac{I_{or}}{I_o} \right)^2 - 1} \quad (71)$$

PIV: after $\omega t = \beta$, $v_T = v_s$. For $\beta \leq 3\pi/2$ the device sees the full negative peak.

$$PIV = V_m \quad (72)$$

For $\beta > 3\pi/2$ the actual reverse peak is $V_m |\sin \beta|$, but the device is always rated for V_m .

Power and input power factor:

$$P = I_{or}^2 R, \quad \text{pf} = \frac{I_{or}^2 R}{V_s I_{or}} = \frac{I_{or} R}{V_s} \quad (73)$$

Circuit turn-off time: $t_c = (2\pi - \beta)/\omega$.

4.9 Special Case: Pure L Load

For $R = 0$, $\theta = \pi/2$ and KVL gives $L di_o/dt = V_m \sin \omega t$. Integrating from α :

$$i_o(\omega t) = \frac{V_m}{\omega L} (\cos \alpha - \cos \omega t) \quad (74)$$

Current is zero again when $\cos \beta = \cos \alpha$:

$$\beta = 2\pi - \alpha, \quad \gamma = 2(\pi - \alpha), \quad V_o = 0 \quad (75)$$

$$I_o = \frac{V_m}{2\pi\omega L} \int_{\alpha}^{2\pi-\alpha} (\cos \alpha - \cos \omega t) d(\omega t) \quad (76)$$

$$= \frac{V_m}{\pi\omega L} [(\pi - \alpha) \cos \alpha + \sin \alpha] \quad (77)$$

5 RL Load with FD

5.1 Operation

At $\omega t = \pi$ the source voltage reverses. The FD becomes forward biased and takes over the load current. T is reverse biased by v_s and turns OFF. The negative part of v_o is removed.

- **Mode 1** ($\alpha \leq \omega t \leq \pi$): T conducts, $v_o = v_s$, $i_s = i_T = i_o$, $i_{FD} = 0$
- **Mode 2** ($\pi < \omega t < 2\pi + \alpha$): FD conducts, $v_o = 0$, $i_{FD} = i_o$, $i_s = 0$

5.2 Average and RMS Output Voltage

The output voltage waveform is the same as that of R load:

$$V_o = \frac{1}{2\pi} \int_{\alpha}^{\pi} V_m \sin \omega t d(\omega t) = \frac{V_m}{2\pi} (1 + \cos \alpha) \quad (78)$$

$$V_{or} = \frac{V_m}{2\sqrt{\pi}} \sqrt{(\pi - \alpha) + \frac{1}{2} \sin 2\alpha} \quad (79)$$

$$\text{FF} = \frac{\sqrt{\pi} \sqrt{(\pi - \alpha) + \frac{1}{2} \sin 2\alpha}}{1 + \cos \alpha}, \quad \text{RF} = \sqrt{\text{FF}^2 - 1} \quad (80)$$

5.3 Load Current Expression

Let $I_\alpha = i_o(\alpha)$ and $I_\pi = i_o(\pi)$.

Mode 1: KVL is $L di_o/dt + R i_o = V_m \sin \omega t$ with $i_o(\alpha) = I_\alpha$.

$$i_o(\omega t) = \frac{V_m}{Z} \sin(\omega t - \theta) + \left[I_\alpha - \frac{V_m}{Z} \sin(\alpha - \theta) \right] e^{-\frac{\omega t - \alpha}{\tan \theta}} \quad (81)$$

At $\omega t = \pi$, using $\sin(\pi - \theta) = \sin \theta$:

$$I_\pi = \frac{V_m}{Z} \sin \theta + \left[I_\alpha - \frac{V_m}{Z} \sin(\alpha - \theta) \right] e^{-\frac{\pi - \alpha}{\tan \theta}} \quad (82)$$

Mode 2: KVL is $L di_o/dt + R i_o = 0$ with $i_o(\pi) = I_\pi$.

$$i_o(\omega t) = I_\pi e^{-\frac{\omega t - \pi}{\tan \theta}} \quad (83)$$

Steady state condition: $i_o(2\pi + \alpha) = I_\alpha$.

$$I_\alpha = I_\pi e^{-\frac{\pi + \alpha}{\tan \theta}} \quad (84)$$

Substituting (84) in (82):

$$I_\pi = \frac{V_m}{Z} \cdot \frac{\sin \theta - \sin(\alpha - \theta) e^{-\frac{\pi - \alpha}{\tan \theta}}}{1 - e^{-\frac{2\pi}{\tan \theta}}} \quad (85)$$

The free wheeling current decays exponentially. For small L , I_α is negligible and the current is treated as discontinuous ($I_\alpha \approx 0$). For large L the current is continuous with small ripple.

5.4 Average Output and Input Currents

Average output current: the net change of inductor current over one period is zero.

$$I_o = \frac{V_o}{R} = \frac{V_m}{2\pi R} (1 + \cos \alpha) \quad (86)$$

Average FD current: integrating the mode 2 current:

$$I_{FD,avg} = \frac{1}{2\pi} \int_{\pi}^{2\pi + \alpha} I_\pi e^{-\frac{\omega t - \pi}{\tan \theta}} d(\omega t) \quad (87)$$

$$= \frac{\tan \theta}{2\pi} I_\pi \left(1 - e^{-\frac{\pi + \alpha}{\tan \theta}} \right) = \frac{\tan \theta}{2\pi} (I_\pi - I_\alpha) \quad (88)$$

Average input (thyristor) current:

$$I_{s,avg} = I_{T,avg} = I_o - I_{FD,avg} = \frac{V_m}{2\pi R} (1 + \cos \alpha) - \frac{\tan \theta}{2\pi} (I_\pi - I_\alpha) \quad (89)$$

5.5 Highly Inductive Load (Ripple Free Current)

For $\omega L \gg R$ the load current is constant at I_o . T carries I_o for an angle $(\pi - \alpha)$ and FD carries I_o for an angle $(\pi + \alpha)$ in every cycle.

$$I_{or} = I_o \quad (90)$$

$$I_{s,avg} = I_{T,avg} = I_o \frac{\pi - \alpha}{2\pi}, \quad I_{s,rms} = I_{T,rms} = I_o \sqrt{\frac{\pi - \alpha}{2\pi}} \quad (91)$$

$$I_{FD,avg} = I_o \frac{\pi + \alpha}{2\pi}, \quad I_{FD,rms} = I_o \sqrt{\frac{\pi + \alpha}{2\pi}} \quad (92)$$

Input power factor:

$$\text{pf} = \frac{V_o I_o}{V_s I_{s,rms}} \quad (93)$$

5.6 PIV

- **Thyristor:** during mode 2, $v_T = v_s - v_o = v_s$. Hence $PIV_T = V_m$
- **FD:** during mode 1, the FD blocks $v_o = v_s$. Hence $PIV_{FD} = V_m$ for $\alpha \leq \pi/2$ and $V_m \sin \alpha$ for $\alpha > \pi/2$

Circuit turn-off time: $t_c = \pi/\omega$.

Advantages of FD:

- Higher average output voltage, as the negative part of v_o is removed
- Improved input power factor
- Smoother load current
- T regains its blocking state at $\omega t = \pi$ and gets full half cycle for turn-off

Example 5.1 (RL Load with FD, Level Current):

A 1-phase half wave controlled rectifier with FD feeds a highly inductive load with $R = 5 \Omega$ from 230 V, 50 Hz supply at $\alpha = 45^\circ$.

Output voltage and current:

$$V_o = \frac{325.27}{2\pi}(1 + \cos 45^\circ) = 51.768 \times 1.7071 = 88.37 \text{ V} \quad (94)$$

$$I_o = \frac{88.37}{5} = 17.67 \text{ A} \quad (95)$$

RMS output voltage, FF and RF:

$$V_{or} = 162.63 \sqrt{\frac{1}{\pi} \left(\pi - \frac{\pi}{4} + \frac{\sin 90^\circ}{2} \right)} = 162.63 \times 0.9535 = 155.07 \text{ V} \quad (96)$$

$$FF = \frac{155.07}{88.37} = 1.755, \quad RF = \sqrt{1.755^2 - 1} = 1.442 \quad (97)$$

Device and input currents:

$$I_{s,avg} = I_{T,avg} = 17.67 \times \frac{135}{360} = 6.63 \text{ A} \quad (98)$$

$$I_{s,rms} = I_{T,rms} = 17.67 \times \sqrt{0.375} = 10.82 \text{ A} \quad (99)$$

$$I_{FD,avg} = 17.67 \times \frac{225}{360} = 11.05 \text{ A} \quad (100)$$

$$I_{FD,rms} = 17.67 \times \sqrt{0.625} = 13.97 \text{ A} \quad (101)$$

Input power factor and PIV:

$$\text{pf} = \frac{88.37 \times 17.67}{230 \times 10.82} = 0.627 \text{ (lag)} \quad (102)$$

$$PIV_T = PIV_{FD} = 325.27 \text{ V} \quad (103)$$

6 RE Load

6.1 RE Load without FD

6.1.1 Operation

The load is a resistance R in series with a DC source E (battery or back EMF). T is forward biased only when $v_s > E$.

$$V_m \sin \delta = E \Rightarrow \delta = \sin^{-1} \left(\frac{E}{V_m} \right) \quad (104)$$

Firing angle range:

$$\delta \leq \alpha \leq \pi - \delta \quad (105)$$

Conduction interval: current flows as long as $v_s > E$.

$$\beta = \pi - \delta, \quad \gamma = \pi - \delta - \alpha \quad (106)$$

- $\alpha \leq \omega t \leq \beta$: T conducts, $v_o = v_s$, $i_o = (V_m \sin \omega t - E)/R$
- Rest of the period: T is OFF, $i_o = 0$, $v_o = E$, $v_T = v_s - E$

6.1.2 Average Output Current

$$I_o = \frac{1}{2\pi R} \int_{\alpha}^{\pi-\delta} (V_m \sin \omega t - E) d(\omega t) \quad (107)$$

$$= \frac{1}{2\pi R} \left[-V_m \cos \omega t - E \omega t \right]_{\alpha}^{\pi-\delta} \quad (108)$$

$$= \frac{1}{2\pi R} [V_m \{\cos \alpha - \cos(\pi - \delta)\} - E(\pi - \delta - \alpha)] \quad (109)$$

$$I_o = \frac{1}{2\pi R} [V_m (\cos \alpha + \cos \delta) - E \gamma] \quad (110)$$

6.1.3 Average Output Voltage

The load terminal voltage is v_s for an angle γ and E for the remaining angle $(2\pi - \gamma)$.

$$V_o = \frac{1}{2\pi} \left[\int_{\alpha}^{\pi-\delta} V_m \sin \omega t d(\omega t) + E(2\pi - \gamma) \right] \quad (111)$$

$$V_o = \frac{1}{2\pi} [V_m (\cos \alpha + \cos \delta) + E(\pi + \delta + \alpha)] \quad (112)$$

Check: $V_o = E + R I_o$ gives the same expression.

6.1.4 RMS Output Current

$$I_{or}^2 = \frac{1}{2\pi R^2} \int_{\alpha}^{\pi-\delta} (V_m \sin \omega t - E)^2 d(\omega t) \quad (113)$$

$$= \frac{1}{2\pi R^2} \int_{\alpha}^{\pi-\delta} (V_m^2 \sin^2 \omega t - 2V_m E \sin \omega t + E^2) d(\omega t) \quad (114)$$

Evaluating term by term, with $V_m^2/2 = V_s^2$ and $\sin 2(\pi - \delta) = -\sin 2\delta$:

$$\int_{\alpha}^{\pi-\delta} V_m^2 \sin^2 \omega t d(\omega t) = V_s^2 \left[\gamma + \frac{1}{2} (\sin 2\alpha + \sin 2\delta) \right] \quad (115)$$

$$\int_{\alpha}^{\pi-\delta} 2V_m E \sin \omega t d(\omega t) = 2V_m E (\cos \alpha + \cos \delta) \quad (116)$$

$$\int_{\alpha}^{\pi-\delta} E^2 d(\omega t) = E^2 \gamma \quad (117)$$

$$I_{or} = \sqrt{\frac{1}{2\pi R^2} \left[(V_s^2 + E^2) \gamma + \frac{V_s^2}{2} (\sin 2\alpha + \sin 2\delta) - 2V_m E (\cos \alpha + \cos \delta) \right]} \quad (118)$$

6.1.5 RMS Output Voltage

$$V_{or}^2 = \frac{1}{2\pi} \left[\int_{\alpha}^{\pi-\delta} V_m^2 \sin^2 \omega t d(\omega t) + E^2 (2\pi - \gamma) \right] \quad (119)$$

$$V_{or} = \sqrt{\frac{1}{2\pi} \left[V_s^2 \left\{ \gamma + \frac{1}{2} (\sin 2\alpha + \sin 2\delta) \right\} + E^2 (\pi + \delta + \alpha) \right]} \quad (120)$$

6.1.6 Input Current, FF, RF and PIV

$$I_{s,avg} = I_o, \quad I_{s,rms} = I_{or} \quad (121)$$

$$FF = \frac{V_{or}}{V_o}, \quad RF = \sqrt{FF^2 - 1}, \quad RF_i = \sqrt{\left(\frac{I_{or}}{I_o} \right)^2 - 1} \quad (122)$$

PIV: in OFF state $v_T = v_s - E$. The most negative value occurs at $\omega t = 3\pi/2$.

$$PIV = V_m + E \quad (123)$$

Power relations:

$$P_E = E I_o \quad (\text{power delivered to the DC source}) \quad (124)$$

$$P_R = I_{or}^2 R \quad (\text{loss in resistance}) \quad (125)$$

$$\text{pf} = \frac{I_{or}^2 R + E I_o}{V_s I_{or}} \quad (126)$$

Circuit turn-off time: T is reverse biased from $\omega t = \pi - \delta$ to $2\pi + \delta$.

$$t_c = \frac{\pi + 2\delta}{\omega} \quad (127)$$

6.2 RE Load with FD

The load terminal voltage is $v_s > E$ during conduction and E otherwise. Thus $v_o > 0$ at every instant. The FD is always reverse biased and never conducts. All expressions of RE load without FD remain valid. The FD must block the peak of v_o , i.e. V_m for $\alpha \leq \pi/2$.

Example 6.1 (RE Load: Battery Charging):

A 120 V battery is charged through $R = 8 \Omega$ from 230 V, 50 Hz supply using a 1-phase half wave controlled rectifier at $\alpha = 40^\circ$.

Angles:

$$\delta = \sin^{-1} \left(\frac{120}{325.27} \right) = 21.65^\circ = 0.3779 \text{ rad} \quad (128)$$

$$\beta = 180^\circ - 21.65^\circ = 158.35^\circ \quad (129)$$

$$\gamma = 158.35^\circ - 40^\circ = 118.35^\circ = 2.0656 \text{ rad} \quad (130)$$

Average charging current:

$$I_o = \frac{1}{2\pi \times 8} [325.27 (0.7660 + 0.9295) - 120 \times 2.0656] \quad (131)$$

$$= \frac{551.50 - 247.87}{50.265} = 6.04 \text{ A} \quad (132)$$

RMS current:

$$I_{or}^2 = \frac{1}{2\pi \times 64} [67300 \times 2.0656 + 26450 (0.9848 + 0.6858) - 78064.8 \times 1.6955] \quad (133)$$

$$= \frac{139014.9 + 44187.6 - 132358.9}{402.12} = 126.44 \quad (134)$$

$$I_{or} = 11.24 \text{ A} \quad (135)$$

Other results:

$$V_o = E + R I_o = 120 + 8 \times 6.04 = 168.32 \text{ V} \quad (136)$$

$$P_E = 120 \times 6.04 = 724.8 \text{ W} \quad (137)$$

$$P_R = 126.44 \times 8 = 1011.5 \text{ W} \quad (138)$$

$$\text{pf} = \frac{724.8 + 1011.5}{230 \times 11.24} = 0.671 \text{ (lag)} \quad (139)$$

$$\text{PIV} = 325.27 + 120 = 445.27 \text{ V} \quad (140)$$

7 RLE Load

7.1 RLE Load without FD

7.1.1 Operation

T can be fired only when $v_s > E$, i.e. $\delta \leq \alpha \leq \pi - \delta$. Due to L , the current continues beyond the instant $v_s = E$ and falls to zero at $\omega t = \beta$.

- $\alpha \leq \omega t \leq \beta$: T conducts, $v_o = v_s$
- $\beta < \omega t < 2\pi + \alpha$: T is OFF, $i_o = 0$, $v_o = E$, $v_T = v_s - E$

7.1.2 Load Current Expression

Step 1: KVL during conduction

$$L \frac{di_o}{dt} + R i_o + E = V_m \sin \omega t \quad (141)$$

Step 2: forced component (by superposition of AC source and DC source)

$$i_{ss} = \frac{V_m}{Z} \sin(\omega t - \theta) - \frac{E}{R} \quad (142)$$

Step 3: complete solution

$$i_o = \frac{V_m}{Z} \sin(\omega t - \theta) - \frac{E}{R} + A e^{-\frac{\omega t - \alpha}{\tan \theta}} \quad (143)$$

Step 4: initial condition $i_o(\alpha) = 0$

$$A = \frac{E}{R} - \frac{V_m}{Z} \sin(\alpha - \theta) \quad (144)$$

Step 5: final expression

$$i_o(\omega t) = \frac{V_m}{Z} \sin(\omega t - \theta) - \frac{E}{R} + \left[\frac{E}{R} - \frac{V_m}{Z} \sin(\alpha - \theta) \right] e^{-\frac{\omega t - \alpha}{\tan \theta}}, \quad \alpha \leq \omega t \leq \beta \quad (145)$$

Extinction angle: putting $i_o(\beta) = 0$:

$$\frac{V_m}{Z} \sin(\beta - \theta) - \frac{E}{R} + \left[\frac{E}{R} - \frac{V_m}{Z} \sin(\alpha - \theta) \right] e^{-\frac{\beta - \alpha}{\tan \theta}} = 0 \quad (146)$$

This transcendental equation is solved numerically for β (Section 12).**7.1.3 Average Output Voltage and Current**

$$V_o = \frac{1}{2\pi} \left[\int_{\alpha}^{\beta} V_m \sin \omega t d(\omega t) + E(2\pi - \gamma) \right] \quad (147)$$

$$V_o = \frac{1}{2\pi} [V_m (\cos \alpha - \cos \beta) + E(2\pi + \alpha - \beta)] \quad (148)$$

Average inductor voltage is zero, so $V_o = E + R I_o$:

$$I_o = \frac{V_o - E}{R} = \frac{1}{2\pi R} [V_m (\cos \alpha - \cos \beta) - E \gamma] \quad (149)$$

7.1.4 RMS Output Voltage and Current

$$V_{or}^2 = \frac{1}{2\pi} \left[\int_{\alpha}^{\beta} V_m^2 \sin^2 \omega t d(\omega t) + E^2(2\pi - \gamma) \right] \quad (150)$$

$$V_{or} = \sqrt{\frac{1}{2\pi} \left[V_s^2 \left\{ \gamma - \frac{1}{2} (\sin 2\beta - \sin 2\alpha) \right\} + E^2(2\pi - \gamma) \right]} \quad (151)$$

$$I_{or} = \sqrt{\frac{1}{2\pi} \int_{\alpha}^{\beta} i_o^2 d(\omega t)} \quad (152)$$

with i_o from (145). This integral is evaluated numerically.**7.1.5 Input Current, FF, RF and PIV**

$$I_{s,avg} = I_o, \quad I_{s,rms} = I_{or} \quad (153)$$

$$FF = \frac{V_{or}}{V_o}, \quad RF = \sqrt{FF^2 - 1} \quad (154)$$

$$PIV = V_m + E \quad (155)$$

Power balance:

$$V_s I_{s,rms} \times \text{pf} = I_{or}^2 R + E I_o \quad (156)$$

7.2 RLE Load with FD

7.2.1 Operation

- **Mode 1** ($\alpha \leq \omega t \leq \pi$): T conducts, $v_o = v_s$
- **Mode 2** ($\pi < \omega t \leq \beta$): FD conducts, $v_o = 0$, energy of L is transferred to R and E
- **Mode 3** ($\beta < \omega t < 2\pi + \alpha$): both devices are OFF, $i_o = 0$, $v_o = E$. This mode exists only for discontinuous conduction

7.2.2 Load Current Expression

Mode 1: with $i_o(\alpha) = I_\alpha$:

$$i_o(\omega t) = \frac{V_m}{Z} \sin(\omega t - \theta) - \frac{E}{R} + \left[I_\alpha + \frac{E}{R} - \frac{V_m}{Z} \sin(\alpha - \theta) \right] e^{-\frac{\omega t - \alpha}{\tan \theta}} \quad (157)$$

$$I_\pi = \frac{V_m}{Z} \sin \theta - \frac{E}{R} + \left[I_\alpha + \frac{E}{R} - \frac{V_m}{Z} \sin(\alpha - \theta) \right] e^{-\frac{\pi - \alpha}{\tan \theta}} \quad (158)$$

Mode 2: KVL is $L di_o/dt + Ri_o + E = 0$ with $i_o(\pi) = I_\pi$.

$$i_o(\omega t) = -\frac{E}{R} + \left(I_\pi + \frac{E}{R} \right) e^{-\frac{\omega t - \pi}{\tan \theta}} \quad (159)$$

Extinction angle (discontinuous conduction): putting $i_o(\beta) = 0$:

$$e^{\frac{\beta - \pi}{\tan \theta}} = 1 + \frac{RI_\pi}{E} \quad (160)$$

$$\beta = \pi + \tan \theta \ln \left(1 + \frac{RI_\pi}{E} \right) \quad (161)$$

Conduction is discontinuous ($I_\alpha = 0$) if $\beta < 2\pi + \alpha$. Otherwise it is continuous and $I_\alpha = i_o(2\pi + \alpha)$ from the mode 2 expression.

7.2.3 Continuous Conduction

$$V_o = \frac{V_m}{2\pi} (1 + \cos \alpha) \quad (162)$$

$$V_{or} = \frac{V_m}{2\sqrt{\pi}} \sqrt{(\pi - \alpha) + \frac{1}{2} \sin 2\alpha} \quad (163)$$

$$I_o = \frac{V_o - E}{R} = \frac{1}{R} \left[\frac{V_m}{2\pi} (1 + \cos \alpha) - E \right] \quad (164)$$

For ripple free current, T carries I_o for $(\pi - \alpha)$ and FD carries I_o for $(\pi + \alpha)$:

$$I_{s,avg} = I_{T,avg} = I_o \frac{\pi - \alpha}{2\pi}, \quad I_{s,rms} = I_{T,rms} = I_o \sqrt{\frac{\pi - \alpha}{2\pi}} \quad (165)$$

$$I_{FD,avg} = I_o \frac{\pi + \alpha}{2\pi}, \quad I_{FD,rms} = I_o \sqrt{\frac{\pi + \alpha}{2\pi}} \quad (166)$$

7.2.4 Discontinuous Conduction

The output voltage is v_s for $(\pi - \alpha)$, zero for $(\beta - \pi)$ and E for $(2\pi + \alpha - \beta)$.

$$V_o = \frac{1}{2\pi} [V_m (1 + \cos \alpha) + E(2\pi + \alpha - \beta)] \quad (167)$$

$$I_o = \frac{V_o - E}{R} = \frac{1}{2\pi R} [V_m (1 + \cos \alpha) - E(\beta - \alpha)] \quad (168)$$

$$V_{or} = \sqrt{\frac{1}{2\pi} \left[V_s^2 \left\{ (\pi - \alpha) + \frac{1}{2} \sin 2\alpha \right\} + E^2 (2\pi + \alpha - \beta) \right]} \quad (169)$$

Average input current: integrating the mode 1 KVL from α to π :

$$V_m (1 + \cos \alpha) = R \int_{\alpha}^{\pi} i_o d(\omega t) + \omega L (I_{\pi} - I_{\alpha}) + E(\pi - \alpha) \quad (170)$$

$$I_{s,avg} = I_{T,avg} = \frac{1}{2\pi R} [V_m (1 + \cos \alpha) - E(\pi - \alpha) - \omega L (I_{\pi} - I_{\alpha})] \quad (171)$$

7.2.5 FF, RF and PIV

$$FF = \frac{V_{or}}{V_o}, \quad RF = \sqrt{FF^2 - 1} \quad (172)$$

- **Thyristor:** $v_T = v_s$ in mode 2 and $v_T = v_s - E$ in mode 3. The design value is $PIV_T = V_m + E$. For continuous conduction the actual reverse peak is V_m .
- **FD:** it blocks v_o . $PIV_{FD} = V_m$ for $\alpha \leq \pi/2$

8 Effect of Source Inductance and Overlap Angle

8.1 Commutation Overlap

A practical source has inductance L_s . Current through L_s cannot change instantaneously. Therefore the transfer of load current between FD and T takes a finite angle μ , called overlap angle. During overlap both T and FD conduct together.

Assumption: load current is constant at I_o (highly inductive load with FD).

8.2 Derivation of Overlap Angle

Step 1: at $\omega t = \alpha$, T is fired. FD is still conducting, so $v_o = 0$. The whole source voltage appears across L_s :

$$L_s \frac{di_T}{dt} = V_m \sin \omega t \Rightarrow \omega L_s di_T = V_m \sin \omega t d(\omega t) \quad (173)$$

Step 2: i_T rises from 0 at $\omega t = \alpha$ to I_o at $\omega t = \alpha + \mu$. At the same time $i_{FD} = I_o - i_T$ falls from I_o to 0.

$$\omega L_s \int_0^{I_o} di_T = \int_{\alpha}^{\alpha+\mu} V_m \sin \omega t d(\omega t) \quad (174)$$

$$\omega L_s I_o = V_m [\cos \alpha - \cos(\alpha + \mu)] \quad (175)$$

Step 3:

$$\cos(\alpha + \mu) = \cos \alpha - \frac{\omega L_s I_o}{V_m} \quad (176)$$

$$\mu = \cos^{-1} \left[\cos \alpha - \frac{\omega L_s I_o}{V_m} \right] - \alpha \quad (177)$$

8.3 Average Output Voltage with Overlap

The output voltage is zero during $\alpha \leq \omega t \leq \alpha + \mu$. It follows v_s only from $\alpha + \mu$ to π .

$$V_o = \frac{1}{2\pi} \int_{\alpha+\mu}^{\pi} V_m \sin \omega t d(\omega t) = \frac{V_m}{2\pi} [1 + \cos(\alpha + \mu)] \quad (178)$$

$$= \frac{V_m}{2\pi} \left[1 + \cos \alpha - \frac{\omega L_s I_o}{V_m} \right] \quad (179)$$

$$V_o = \frac{V_m}{2\pi} (1 + \cos \alpha) - \frac{\omega L_s I_o}{2\pi} \quad (180)$$

Since $\omega = 2\pi f$:

$$V_o = \frac{V_m}{2\pi} (1 + \cos \alpha) - f L_s I_o \quad (181)$$

Voltage drop due to overlap:

$$\Delta V_o = f L_s I_o \quad (182)$$

This drop behaves like a loss-free equivalent resistance $R_{eq} = f L_s$ in series with the DC side.

8.4 Overlap at the End of Conduction

At $\omega t = \pi$ the FD starts to conduct and i_T falls from I_o to 0 over an angle μ_2 . During this interval $v_o = 0$, which is the same as the ideal case. Hence no additional drop appears in V_o .

$$\omega L_s \int_{I_o}^0 di_T = \int_{\pi}^{\pi+\mu_2} V_m \sin \omega t d(\omega t) = -V_m (1 - \cos \mu_2) \quad (183)$$

$$\cos \mu_2 = 1 - \frac{\omega L_s I_o}{V_m} \quad (184)$$

Example 8.1 (Overlap Angle):

A 1-phase half wave controlled rectifier with FD feeds a level load current of 15 A from 230 V, 50 Hz supply. The source inductance is 2 mH and $\alpha = 30^\circ$.

Source reactance:

$$\omega L_s = 2\pi \times 50 \times 0.002 = 0.6283 \Omega \quad (185)$$

Overlap angle:

$$\cos(\alpha + \mu) = \cos 30^\circ - \frac{0.6283 \times 15}{325.27} = 0.8660 - 0.0290 = 0.8370 \quad (186)$$

$$\alpha + \mu = 33.17^\circ \Rightarrow \mu = 3.17^\circ \quad (187)$$

Average output voltage:

$$V_o(\text{ideal}) = 51.768 (1 + 0.8660) = 96.60 \text{ V} \quad (188)$$

$$\Delta V_o = 50 \times 0.002 \times 15 = 1.50 \text{ V} \quad (189)$$

$$V_o = 96.60 - 1.50 = 95.10 \text{ V} \quad (190)$$

9 Summary of Derived Expressions

9.1 Average Output Voltage and Current

Table 2: Extinction angle, average output voltage and average output current

Load	β	V_o	I_o
R (without or with FD)	π	$\frac{V_m}{2\pi}(1 + \cos \alpha)$	$\frac{V_o}{R}$
RL without FD	from (49)	$\frac{V_m}{2\pi}(\cos \alpha - \cos \beta)$	$\frac{V_o}{R}$
RL with FD	FD conducts after π	$\frac{V_m}{2\pi}(1 + \cos \alpha)$	$\frac{V_o}{R}$
RE (without or with FD)	$\pi - \delta$	$\frac{1}{2\pi} [V_m(\cos \alpha + \cos \delta) + E(\pi + \delta + \alpha)]$	$\frac{V_o - E}{R}$
RLE without FD	numerical	$\frac{1}{2\pi} [V_m(\cos \alpha - \cos \beta) + E(2\pi + \alpha - \beta)]$	$\frac{V_o - E}{R}$
RLE with FD (continuous)	not applicable	$\frac{V_m}{2\pi}(1 + \cos \alpha)$	$\frac{V_o - E}{R}$
RLE with FD (discontinuous)	$\pi + \tan^{-1} \left(\frac{R I_o}{E} \right)$	$\frac{1}{2\pi} [V_m(1 + \cos \alpha) + E(2\pi + \alpha - \beta)]$	$\frac{V_o - E}{R}$

9.2 RMS Output Voltage and PIV

Table 3: RMS output voltage and thyristor PIV

Load	V_{or}^2	Thyristor PIV
R (without or with FD)	$\frac{V_m^2}{4\pi} [(\pi - \alpha) + \frac{1}{2} \sin 2\alpha]$	V_m
RL without FD	$\frac{V_m^2}{4\pi} [(\beta - \alpha) - \frac{1}{2}(\sin 2\beta - \sin 2\alpha)]$	V_m
RL with FD	$\frac{V_m^2}{4\pi} [(\pi - \alpha) + \frac{1}{2} \sin 2\alpha]$	V_m
RE (without or with FD)	$\frac{1}{2\pi} [V_s^2 \{ \gamma + \frac{1}{2}(\sin 2\alpha + \sin 2\delta) \} + E^2(\pi + \delta + \alpha)]$	$V_m + E$
RLE without FD	$\frac{1}{2\pi} [V_s^2 \{ \gamma - \frac{1}{2}(\sin 2\beta - \sin 2\alpha) \} + E^2(2\pi - \gamma)]$	$V_m + E$

Table 3 (continued from previous page)

Load	V_{or}^2	Thyristor PIV
RLE with FD (continuous)	$\frac{V_m^2}{4\pi} [(\pi - \alpha) + \frac{1}{2} \sin 2\alpha]$	V_m (design: $V_m + E$)
RLE with FD (discontinuous)	$\frac{1}{2\pi} [V_s^2 \{(\pi - \alpha) + \frac{1}{2} \sin 2\alpha\} + E^2(2\pi + \alpha - \beta)]$	$V_m + E$

9.3 Input Current, FF and RF

- Without FD (all loads): $I_{s,avg} = I_o$ and $I_{s,rms} = I_{or}$
- With FD, level load current: $I_{s,avg} = I_o \frac{\pi - \alpha}{2\pi}$ and $I_{s,rms} = I_o \sqrt{\frac{\pi - \alpha}{2\pi}}$
- All cases: $FF = \frac{V_{or}}{V_o}$ and $RF = \sqrt{FF^2 - 1}$
- With source inductance and FD: $V_o = \frac{V_m}{2\pi}(1 + \cos \alpha) - fL_s I_o$

10 Practice Problems

10.1 R Load

Problem 10.1 (R Load at $\alpha = 90^\circ$):

A 1-phase half wave controlled rectifier feeds $R = 20 \Omega$ from 230 V, 50 Hz supply. For $\alpha = 90^\circ$ determine V_o , V_{or} , I_o , I_{or} , FF and RF.

Solution:

Average values:

$$V_o = \frac{325.27}{2\pi}(1 + \cos 90^\circ) = 51.77 \text{ V} \quad (191)$$

$$I_o = \frac{51.77}{20} = 2.588 \text{ A} \quad (192)$$

RMS values:

$$V_{or} = \frac{325.27}{2} \sqrt{\frac{1}{\pi} \left(\pi - \frac{\pi}{2} + \frac{\sin 180^\circ}{2} \right)} = 162.63 \times 0.7071 = 115.0 \text{ V} \quad (193)$$

$$I_{or} = \frac{115.0}{20} = 5.75 \text{ A} \quad (194)$$

Form factor and ripple factor:

$$FF = \frac{115.0}{51.77} = 2.221 \quad (195)$$

$$RF = \sqrt{2.221^2 - 1} = 1.984 \quad (196)$$

Conclusion: FF rises from 1.571 at $\alpha = 0$ to 2.221 at $\alpha = 90^\circ$. Ripple increases with firing angle.

10.2 Pure L Load

Problem 10.2 (Pure Inductive Load):

A 1-phase half wave controlled rectifier feeds a pure inductance $L = 0.1$ H from 230 V, 50 Hz supply at $\alpha = 60^\circ$. Determine β , γ , peak current and average current.

Solution:

Angles:

$$\beta = 2\pi - \alpha = 360^\circ - 60^\circ = 300^\circ \quad (197)$$

$$\gamma = 300^\circ - 60^\circ = 240^\circ \quad (198)$$

Reactance:

$$\omega L = 2\pi \times 50 \times 0.1 = 31.416 \, \Omega \quad (199)$$

Peak current (at $\omega t = \pi$):

$$I_{\text{peak}} = \frac{V_m}{\omega L} (\cos \alpha + 1) = \frac{325.27}{31.416} \times 1.5 = 15.53 \, \text{A} \quad (200)$$

Average current:

$$I_o = \frac{V_m}{\pi \omega L} [(\pi - \alpha) \cos \alpha + \sin \alpha] \quad (201)$$

$$= \frac{325.27}{\pi \times 31.416} [2.0944 \times 0.5 + 0.8660] = 3.2957 \times 1.9132 = 6.31 \, \text{A} \quad (202)$$

Conclusion: $V_o = 0$ although $I_o \neq 0$, since a pure inductance absorbs no average power.

10.3 RLE Load with FD

Problem 10.3 (RLE Load with FD, Continuous Conduction):

A 1-phase half wave controlled rectifier with FD feeds an RLE load with $R = 2 \, \Omega$, $E = 40$ V and very large L from 230 V, 50 Hz supply at $\alpha = 30^\circ$. Determine V_o , I_o , average currents of T and FD, power delivered to E and thyristor voltage rating.

Solution:

Check of firing:

$$\delta = \sin^{-1} \left(\frac{40}{325.27} \right) = 7.06^\circ < \alpha \quad \checkmark \quad (203)$$

Average output voltage and current:

$$V_o = \frac{325.27}{2\pi} (1 + \cos 30^\circ) = 51.768 \times 1.8660 = 96.60 \, \text{V} \quad (204)$$

$$I_o = \frac{96.60 - 40}{2} = 28.30 \, \text{A} \quad (205)$$

Device currents:

$$I_{T,\text{avg}} = I_{s,\text{avg}} = 28.30 \times \frac{150}{360} = 11.79 \, \text{A} \quad (206)$$

$$I_{FD,\text{avg}} = 28.30 \times \frac{210}{360} = 16.51 \, \text{A} \quad (207)$$

Power delivered to E :

$$P_E = 40 \times 28.30 = 1132 \text{ W} \quad (208)$$

Thyristor voltage rating:

$$\text{PIV}_T(\text{design}) = V_m + E = 325.27 + 40 = 365.27 \text{ V} \quad (209)$$

Conclusion: FD carries more average current than T, since it conducts for 210° against 150° of T.

11 Summary and Key Learning Outcomes

11.1 Fundamental Concepts Mastered

Through the present note, students have developed thorough understanding of:

1. **Phase control:** variation of V_o from V_m/π to zero by changing α from 0 to π
2. **Effect of load inductance:** extension of conduction up to $\beta > \pi$ and reduction of V_o
3. **Effect of FD:** removal of negative output voltage, so that V_o is restored to the R load value
4. **Effect of load EMF:** restricted firing range $\delta \leq \alpha \leq \pi - \delta$ and increase of PIV to $V_m + E$
5. **Effect of source inductance:** overlap angle μ and voltage drop $fL_s I_o$
6. **Quality of output:** FF and RF, both of which increase with α

11.2 Key Mathematical Relationships

$$\text{R load: } V_o = \frac{V_m}{2\pi} (1 + \cos \alpha) \quad (210)$$

$$\text{RL load without FD: } V_o = \frac{V_m}{2\pi} (\cos \alpha - \cos \beta) \quad (211)$$

$$\text{RE load: } I_o = \frac{1}{2\pi R} [V_m (\cos \alpha + \cos \delta) - E\gamma] \quad (212)$$

$$\text{RLE load without FD: } I_o = \frac{1}{2\pi R} [V_m (\cos \alpha - \cos \beta) - E\gamma] \quad (213)$$

$$\text{Ripple factor: } \text{RF} = \sqrt{\text{FF}^2 - 1} \quad (214)$$

$$\text{Overlap: } \cos(\alpha + \mu) = \cos \alpha - \frac{\omega L_s I_o}{V_m} \quad (215)$$

Definition 11.1 (Module Integration): *The half wave circuit is the building block of all phase controlled converters. The same two steps, KVL for each conducting interval and integration over one period, are used for full wave, semi converter and three-phase converters in the subsequent lectures.*

12 Numerical Determination of Extinction Angle β

The extinction angle β is required in every expression of RL and RLE loads. This section shows when β has a direct formula and how it is computed when it is not given in the problem.

12.1 Cases with Direct Formula

Load	Extinction angle
R (without or with FD)	$\beta = \pi$
RE (without or with FD)	$\beta = \pi - \delta, \quad \delta = \sin^{-1}(E/V_m)$
Pure L without FD	$\beta = 2\pi - \alpha$
RLE with FD (discontinuous)	$\beta = \pi + \tan \theta \ln \left(1 + \frac{R I_\pi}{E} \right)$

12.2 Cases without Direct Formula

For RL and RLE loads without FD, β appears inside a sine term and inside an exponential term. No closed form solution exists. In numerical problems one of the following three routes is used:

1. **β or γ is given.** Then $\beta = \alpha + \gamma$ is used directly
2. **V_o is given.** For RL load, $\cos \beta = \cos \alpha - \frac{2\pi V_o}{V_m}$
3. **Nothing is given.** β is computed by iteration, say Newton-Raphson (NR) method

12.3 Newton-Raphson Formulation for RL Load without FD

Step 1: write the extinction equation (49) as $f(\beta) = 0$.

$$f(\beta) = \sin(\beta - \theta) - K e^{-\frac{\beta - \alpha}{\tan \theta}}, \quad K = \sin(\alpha - \theta) \quad (216)$$

Step 2: differentiate with respect to β .

$$f'(\beta) = \cos(\beta - \theta) + \frac{K}{\tan \theta} e^{-\frac{\beta - \alpha}{\tan \theta}} \quad (217)$$

Step 3: NR update.

$$\beta_{k+1} = \beta_k - \frac{f(\beta_k)}{f'(\beta_k)} \quad (218)$$

Step 4: initial guess. The exponential term is small near extinction. Neglecting it gives $\sin(\beta - \theta) = 0$:

$$\beta_0 = \pi + \theta \quad (219)$$

Step 5: stopping rule. Stop when $|\beta_{k+1} - \beta_k| < 10^{-4}$ rad.

Caution: $\omega t = \alpha$ is also a root, since $f(\alpha) = K - K = 0$. This is the starting instant of conduction and not the extinction instant. The initial guess must therefore be taken beyond π . All angles must be in radian.

12.4 Fixed Point Alternative

Since $\beta - \theta$ lies near π , equation (49) can be rearranged as:

$$\beta_{k+1} = \pi + \theta - \sin^{-1} \left[K e^{-\frac{\beta_k - \alpha}{\tan \theta}} \right] \quad (220)$$

Starting from $\beta_0 = \pi + \theta$, this converges in 2 to 3 steps and needs no derivative.

12.5 Quick Checks on β

- $\alpha = \theta$: $K = 0$, the transient term vanishes and $\beta = \pi + \theta$ exactly
- $\alpha < \theta$: $K < 0$ and β is slightly above $\pi + \theta$
- $\alpha > \theta$: $K > 0$ and β is slightly below $\pi + \theta$
- For RL load without FD, $\pi < \beta \leq 2\pi - \alpha$

Example 12.1 (RL Load without FD: β by NR Method):

A 1-phase half wave controlled rectifier feeds an RL load with $R = 10 \Omega$ and $L = 31.83 \text{ mH}$ from 230 V, 50 Hz supply at $\alpha = 30^\circ$. No FD is used and β is not given. Find β , γ , V_o and I_o .

Load data:

$$\omega L = 2\pi \times 50 \times 0.03183 = 10 \Omega \quad (221)$$

$$\theta = \tan^{-1}(10/10) = 45^\circ = 0.785398 \text{ rad}, \quad \tan \theta = 1 \quad (222)$$

$$\alpha = 30^\circ = 0.523599 \text{ rad} \quad (223)$$

$$K = \sin(30^\circ - 45^\circ) = -0.258819 \quad (224)$$

Functions:

$$f(\beta) = \sin(\beta - 0.785398) + 0.258819 e^{-(\beta - 0.523599)} \quad (225)$$

$$f'(\beta) = \cos(\beta - 0.785398) - 0.258819 e^{-(\beta - 0.523599)} \quad (226)$$

Initial guess:

$$\beta_0 = \pi + \theta = 3.926991 \text{ rad} \quad (227)$$

Iteration 1:

$$e^{-(3.926991 - 0.523599)} = e^{-3.403392} = 0.033260 \quad (228)$$

$$f(\beta_0) = \sin(\pi) + 0.258819 \times 0.033260 = 0.008608 \quad (229)$$

$$f'(\beta_0) = \cos(\pi) - 0.008608 = -1.008608 \quad (230)$$

$$\beta_1 = 3.926991 - \frac{0.008608}{-1.008608} = 3.935526 \text{ rad} \quad (231)$$

Iteration 2:

$$e^{-(3.935526 - 0.523599)} = e^{-3.411927} = 0.032977 \quad (232)$$

$$f(\beta_1) = \sin(3.150128) + 0.258819 \times 0.032977 = -0.008535 + 0.008535 \approx 0 \quad (233)$$

$$\beta_2 = \beta_1 = 3.9355 \text{ rad} \quad (234)$$

Result:

$$\beta = 3.9355 \text{ rad} = 225.49^\circ \quad (235)$$

$$\gamma = 225.49^\circ - 30^\circ = 195.49^\circ \quad (236)$$

Since $\alpha < \theta$, β is slightly above $\pi + \theta = 225^\circ$, as expected.

Average output voltage and current:

$$V_o = \frac{325.27}{2\pi} (\cos 30^\circ - \cos 225.49^\circ) = 51.768 (0.8660 + 0.7010) = 81.12 \text{ V} \quad (237)$$

$$I_o = \frac{81.12}{10} = 8.11 \text{ A} \quad (238)$$

12.6 Newton-Raphson Formulation for RLE Load without FD

Step 1: from (145), with $B = \frac{E}{R} - \frac{V_m}{Z} \sin(\alpha - \theta)$:

$$g(\beta) = \frac{V_m}{Z} \sin(\beta - \theta) - \frac{E}{R} + B e^{-\frac{\beta - \alpha}{\tan \theta}} \quad (239)$$

Step 2:

$$g'(\beta) = \frac{V_m}{Z} \cos(\beta - \theta) - \frac{B}{\tan \theta} e^{-\frac{\beta - \alpha}{\tan \theta}} \quad (240)$$

Step 3:

$$\beta_{k+1} = \beta_k - \frac{g(\beta_k)}{g'(\beta_k)} \quad (241)$$

Step 4: initial guess. Neglecting the exponential term gives $\frac{V_m}{Z} \sin(\beta - \theta) = \frac{E}{R}$. Since $\frac{Z}{R} = \frac{1}{\cos \theta}$, this is $\sin(\beta - \theta) = \frac{E}{V_m \cos \theta}$. Taking the root on the falling part of the sine wave:

$$\beta_0 = \pi + \theta - \sin^{-1} \left(\frac{E}{V_m \cos \theta} \right) \quad (242)$$

If $E > V_m \cos \theta$, the inverse sine does not exist and $\beta_0 = \pi$ is used.

Example 12.2 (RLE Load without FD: β by NR Method):

A 1-phase half wave controlled rectifier feeds an RLE load with $R = 10 \Omega$, $\omega L = 10 \Omega$ and $E = 100 \text{ V}$ from 230 V, 50 Hz supply at $\alpha = 45^\circ$. No FD is used and β is not given. Find β , γ , I_o and V_o .

Load data:

$$\theta = 45^\circ = 0.785398 \text{ rad}, \quad \tan \theta = 1, \quad Z = 14.142 \Omega \quad (243)$$

$$\frac{V_m}{Z} = \frac{325.27}{14.142} = 23 \text{ A}, \quad \frac{E}{R} = 10 \text{ A} \quad (244)$$

$$\delta = \sin^{-1} \left(\frac{100}{325.27} \right) = 17.90^\circ < \alpha \quad \checkmark \quad (245)$$

$$B = 10 - 23 \sin(45^\circ - 45^\circ) = 10 \text{ A} \quad (246)$$

Functions: let $u = \beta - \theta$. In this example $\alpha = \theta$, so $\beta - \alpha$ is also equal to u .

$$g(u) = 23 \sin u - 10 + 10 e^{-u} \quad (247)$$

$$g'(u) = 23 \cos u - 10 e^{-u} \quad (248)$$

Initial guess:

$$u_0 = \pi - \sin^{-1} \left(\frac{10}{23} \right) = 3.14159 - 0.44979 = 2.69180 \text{ rad} \quad (249)$$

Iteration 1:

$$g(u_0) = 23 \times 0.43478 - 10 + 10 \times 0.067759 = 0.67759 \quad (250)$$

$$g'(u_0) = 23 \times (-0.90054) - 0.67759 = -21.3900 \quad (251)$$

$$u_1 = 2.69180 - \frac{0.67759}{-21.3900} = 2.72348 \text{ rad} \quad (252)$$

Iteration 2:

$$g(u_1) = 23 \times 0.406037 - 10 + 10 \times 0.065646 = -0.00468 \quad (253)$$

$$g'(u_1) = 23 \times (-0.913857) - 0.65646 = -21.6752 \quad (254)$$

$$u_2 = 2.72348 - \frac{-0.00468}{-21.6752} = 2.72326 \text{ rad} \quad (255)$$

Iteration 3: $|u_2 - u_1| = 0.00022 \text{ rad}$ and $g(u_2) \approx 0$. The solution has converged.

Result:

$$\beta = u + \theta = 2.7233 + 0.7854 = 3.5087 \text{ rad} = 201.03^\circ \quad (256)$$

$$\gamma = 201.03^\circ - 45^\circ = 156.03^\circ = 2.7233 \text{ rad} \quad (257)$$

Average output current and voltage:

$$I_o = \frac{1}{2\pi \times 10} [325.27 (\cos 45^\circ - \cos 201.03^\circ) - 100 \times 2.7233] \quad (258)$$

$$= \frac{325.27 \times 1.6405 - 272.33}{62.832} = \frac{533.60 - 272.33}{62.832} = 4.16 \text{ A} \quad (259)$$

$$V_o = E + RI_o = 100 + 10 \times 4.16 = 141.6 \text{ V} \quad (260)$$

Remark: $u = 0$ (i.e. $\beta = \alpha$) also satisfies $g(u) = 0$. It is the turn-on instant and must be rejected.

13 Nomenclature and Abbreviations

13.1 Symbols

Symbol	Description
v_s	Instantaneous source voltage (V)
V_s	RMS source voltage (V)
V_m	Peak source voltage, $\sqrt{2}V_s$ (V)
v_o, i_o	Instantaneous output (load) voltage (V) and current (A)
V_o, I_o	Average output voltage (V) and current (A)
V_{or}, I_{or}	RMS output voltage (V) and current (A)
V_{ac}	RMS value of ripple component of output voltage (V)
i_s	Instantaneous source (input) current (A)
$I_{s,avg}, I_{s,rms}$	Average and RMS input current (A)
i_T, i_{FD}	Instantaneous thyristor and free wheeling diode current (A)
$I_{T,avg}, I_{T,rms}$	Average and RMS thyristor current (A)
$I_{FD,avg}, I_{FD,rms}$	Average and RMS free wheeling diode current (A)
v_T	Voltage across thyristor (V)
I_α, I_π	Load current at $\omega t = \alpha$ and $\omega t = \pi$ (A)
R	Load resistance (Ω)
L	Load inductance (H)
E	Load EMF or battery voltage (V)
L_s	Source inductance (H)
Z	Load impedance, $\sqrt{R^2 + (\omega L)^2}$ (Ω)
ω	Angular frequency of supply, $2\pi f$ (rad/s)
f	Supply frequency (Hz)
α	Firing angle (rad)

Symbol	Description
β	Extinction angle (rad)
γ	Conduction angle, $\beta - \alpha$ (rad)
θ	Load impedance angle, $\tan^{-1}(\omega L/R)$ (rad)
μ	Overlap angle (rad)
δ	Angle at which $v_s = E$, $\sin^{-1}(E/V_m)$ (rad)
A, K, B	Constants of integration and substitution
$f(\beta), g(\beta)$	Extinction functions of RL and RLE loads for NR method
β_k, u_k	Value of β and of $u = \beta - \theta$ at k -th iteration (rad)
J_1, J_2, J_3	Intermediate integrals of RMS current derivation
P, P_E, P_R	Load power, power to E , loss in R (W)
η	Rectification efficiency
t_c	Circuit turn-off time (s)

13.2 Abbreviations

Abbreviation	Full Form
AC	Alternating Current
DC	Direct Current
EMF	Electromotive Force
FD	Free Wheeling Diode
FF	Form Factor
KVL	Kirchhoff's Voltage Law
NR	Newton-Raphson
ODE	Ordinary Differential Equation
pf	Power Factor
PIV	Peak Inverse Voltage
R	Resistive Load
RE	Resistive Load with EMF
RF	Ripple Factor
RL	Resistive Inductive Load
RLE	Resistive Inductive Load with EMF
RMS	Root Mean Square
SCR	Silicon Controlled Rectifier
T	Thyristor

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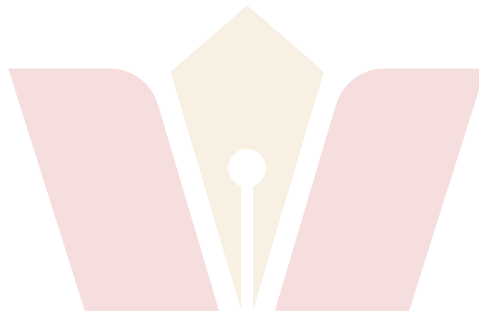
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